

Product Name: **METHYL ACRYLATE**

1. Chemical product and company identification

Product name: **METHYL ACRYLATE**

Name of the supplier: NIPPON SHOKUBAI CO., LTD.
 Address: Kogin Bldg., 4-1-1 Koraibashi, Chuo-ku, Osaka 541-0043 Japan
 Charge section: Responsible Care Division
 (TEL:81-6-6223-9166, FAX:81-6-6202-1766, E-MAIL:MSDS_NSCL@shokubai.co.jp)
 Emergency phone number: Himeji Plant Environmental & Safety Department
 (TEL:81-79-273-1139, FAX:81-79-274-3723)
 Recommended use: Acrylic fibers, Fiber treatment, Paints, Paper treatment

2. Hazard identification

GHS classification of product	[Category]
Flammable liquids	Category 2
Pyrophoric liquids	Not classified
Acute toxicity (oral)	Category 3
Acute toxicity (dermal)	Category 4
Acute toxicity (inhalation:vapour)	Category 3
Skin corrosion/irritation	Category 1
Serious eye damage/eye irritation	Category 1
Skin sensitization	Category 1A
Germ cell mutagenicity	Not classified
Carcinogenicity	Category 1B
Reproductive toxicity	Not classified
Specific target organ systemic toxicity following single exposure	Category 1
Specific target organ systemic toxicity following repeated exposure	Category 1
Acute hazards to the aquatic environment	Category 2
Chronic hazards to the aquatic environment	Category 2

GHS label elements

Pictograms:



Signal word: **Danger**
 Hazard statement: **Highly flammable liquid and vapour**
Toxic if swallowed
Harmful in contact with skin

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Toxic if inhaled
Causes severe skin burns and eye damage
Causes serious eye damage
May cause an allergic skin reaction
May cause cancer
Causes damage to organs(respiratory apparatus)
Causes damage to organs(respiratory apparatus) through prolonged or repeated exposure
Toxic to aquatic life
Toxic to aquatic life with long lasting effects

Precautionary statements:

Prevention Obtain special instructions before use.

Do not handle until all safety precautions have been read and understood.

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

Keep container tightly closed.

Ground and bond container and receiving equipment.

Use explosion-proof [electrical/ventilating/lighting] equipment.

Use non-sparking tools.

Take action to prevent static discharges.

Do not breathe dust/fume/gas/mist/vapours/spray.

Wash hands and face thoroughly after handling.

Do not eat, drink or smoke when using this product.

Use only outdoors or in a well-ventilated area.

Contaminated work clothing should not be allowed out of the workplace.

Avoid release to the environment.

Wear protective gloves/protective clothing/eye protection/face protection.

Response Immediately call a POISON CENTER/doctor.

Get medical advice/attention if you feel unwell.

Rinse mouth.

Take off contaminated clothing and wash it before reuse.

Collect spillage.

IF SWALLOWED: Immediately call a POISON CENTER/doctor.

IF ON SKIN: Wash with plenty of water.

IF INHALED: Remove person to fresh air and keep comfortable for breathing.

IF exposed or concerned: Get medical advice/attention.

If skin irritation or a rash occurs: Get medical advice/attention.

In case of fire : Use appropriate extinguishing media to extinguish.

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

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IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Storage Store locked up.

Store in a well ventilated place. Keep container tightly closed.

Store in a well ventilated place. Keep cool.

Disposal Dispose of contents/container in accordance with local/regional/national/international regulation.

Other hazards:

The product can be charged with static electricity.

Even though a polymerization inhibitor is added, polymerization can be initiated by heating, direct sunlight, peroxides, and iron rust. Progress of polymerization can cause temperature increase and accelerated vapor pressure rise, resulting in explosion.

3. Composition/information on ingredients

Substance or mixture:

Substance

Chemical name	Content wt. %	CAS number
Methyl acrylate(Inhibited)	100	96-33-3

4. First-aid measures

Inhalation:

Remove person to fresh air and keep comfortable for breathing.

Get medical advice/attention.

Skin contact:

Immediately wipe away the attached material with cloths and the like rapidly.

Wash with plenty of soap and water.

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].

Immediately call a POISON CENTER/doctor.

Eye contact:

Wash the eyes with clean water for at least several minutes (desirably 15 minutes or longer), (Remove the contact lenses if possible.) and get medical attention by an eye specialist immediately. During washing, fully open the eyelid with fingers and wash in such a way as to let water reach to every corner of the eyeball and eyelid.

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Ingestion: Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Get medical attention immediately.

5. Fire-fighting measures

Suitable extinguishing media: Small Fire: Chemical powder extinguisher, alcohol resistant foam fire extinguishing media, carbon dioxide or water spray.
Large Fire: Alcohol resistant foam fire extinguishing media, water spray or fog.

Unsuitable extinguishing media: Direct straight streams (may spread a fire)

Specific hazards of fire-fighting: **HIGHLY FLAMMABLE:** Will be easily ignited by heat, sparks or flames.
Containers may explode when heated. Fire will produce irritating, corrosive and/or toxic gases. May polymerize explosively when heated or involved in a fire. Vapor explosion hazard indoors, outdoors or in sewers.

Specific extinguishing methods: Keep upwind of fire.
This product has a very low flash point: Use of water spray at a large fire, when another fire extinguishing media may be inefficient.
Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Cool containers with flooding quantities of water until well after fire is out. For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.

Special protective actions
for fire-fighters: In case of fire fighting, wear positive pressure self-contained breathing apparatus (SCBA) and chemical protective clothing. Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
Structural firefighters' protective clothing will only provide limited protection.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures: Promptly remove possible ignition sources from the vicinity.
Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel.

Environmental precautions: Do not flush to sewer or allow to enter waterways.
This product has a strong bad smell and irritation. Therefore, take proper measures like informing the people in the neighboring area of the leakage occurrence.

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Surround this product with sandbags and earth/sand and cover it with an antistatic sheet to prevent diffusion (of smell).

Methods and materials

for containment and cleaning up: In the case of small amount, hydrolyze it with 5 - 10% sodium hydroxide solution, neutralize, and wash with water. The waste water should be properly treated (incineration, activated sludge process). In the case of large amount, surround it with earth/sand, sandbags and others, cover it with oil-resistant antistatic sheet to prevent generation of vapor, and recover it into proper containers like drums. Then, treat it as in the case of small amount.

7. Handling and storage

Handling

Technical measures: This product can be charged with static electricity. Take countermeasures for static electricity removal (grounding, others). Wear antistatic clothes and antistatic shoes to prevent human body electrification.

Use explosion-proof [electrical/ventilating/lighting] equipment.

Use non-sparking tools.

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

Do not handle the containers harshly in such a way as to make them tumbled, dropped, given a shock, or dragged.

Local/total ventilation: Use the ventilation equipment described in Section 8.

Precautions for safe handling: Not especially.

Contact avoidance: Refer to the information in section 10 (Stability and reactivity).

Hygiene measures: During handling, be sure to wear proper protective equipment (refer to the section 8).

Storage

Conditions for safe storage: Keep away from heat.

Keep away from sources of ignition - No smoking.

Protect from sunlight. Store in a well-ventilated place.

When storing this product in a tank, store it at as-low-as-possible temperature of 30°C or lower and control to keep the oxygen concentration of the gaseous phase at 5 - 8%. (Storing at oxygen concentration of lower than 5% can cause polymerization.)

Safe packaging material: Store in tightly closed container.

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Stainless steel, steel or polyethylene container

8. Exposure controls/personal protection

Permissible concentration

Chemical name	ACGIH		U. S. -OSHA	
	TLV-TWA	TLV-STEL	PEL-TWA	PEL-STEL
Methyl acrylate(Inhibited)	2ppm Skin	Not established	10ppm Skin 35mg/m3 Skin	Not established

Appropriate engineering controls: Install eye and body washing facilities near the handling place.

Use closed equipment/machinery or use local ventilation equipment.

Personal protective equipment

Respiratory protection: Gas mask

Hand protection: Organic solvent impermeable protective gloves (Antistatic ones are desirable.)

Eye/face protection: Protective glasses, goggle, protective face shield

Skin and body protection: Organic solvent impermeable protective clothes, protective shoes (Antistatic ones are desirable.)

9. Physical and chemical properties

[Chemical product]

Physical state: Liquid
 Form: Liquid
 Colour: Colorless
 Odour: Strong ester odor
 Odour threshold: 0.003-0.02 (ppm)

Melting point
 /freezing point: -76.5(°C)

Boiling point or
 initial boiling point
 and boiling range: 80.3(°C)

Flammability: No data

Lower and upper
 explosion limit
 /flammability limit: 1.7-25(Vol%)

Flashpoint: -1.5(°C)Tag Closed Cup
 -2.8(°C)Open Cup

Auto-ignition
 temperature: 420(°C)

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Decomposition
temperature: No data

pH: 6.1(20°C Saturated aqueous solution)

Viscosity : 0.48(mPa·s) (25°C)

Kinematic Viscosity : No data

Solubility: water : 5.2(mass%) (20°C)

Partition coefficient
n-octanol/water
(log value): No data

Vapour pressure: 9145.9(Pa) (20°C)

Density and/or
relative density: 0.9565(20°C)

Relative vapour density: 2.97

Evaporation rate: 3.5

Particle
characteristics: No data

Other data: Conductivity:1.1E-07 S/m
Heat accumulation storage test(UN recommendations): SAPT>50°C
(SAPT:Self-accelerating polymerization temperature)

10. Stability and reactivity

Reactivity: Refer to the information in "Possibility of hazardous reactions".

Chemical stability: Even though a polymerization inhibitor is added, heating, direct sunlight, peroxides, and iron rust can cause polymerization.

Possibility of
hazardous reactions: When storing at low-oxygen condition of oxygen concentration of lower than 5% at the gaseous phase, polymerization can occur.

Conditions to avoid: Heating, contact with incompatible substances, ignition sources

Incompatible materials: Strong oxidizing substances, peroxides.

Hazardous
decomposition products: No data available.

11. Toxicological information

[Chemical product]

Acute toxicity, oral: Category 3
No test data on the product

Acute toxicity, oral: Category 3
LD50: 277-765 mg/kg[rat]

Acute toxicity, dermal: Category 4
LD50: 1243 mg/kg[rat]
LD50: 1243 mg/kg[rabbit]

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Acute toxicity, inhalation:	Not classified (Gas) Category 3 (Vapour) LC50: 750-1850 ppm[rat] Classification not possible (Dust/Mist)
Skin corrosion/irritation:	Category 1 Corrosive [rabbit]
Serious eye damage/irritation:	Category 1 Severe [rabbit]
Sensitization, respiratory:	Classification not possible
Sensitization, skin:	Category 1A Japan Society for Occupational Health Skin Sensitization Group 2
Germ cell mutagenicity:	Not classified Negative
Carcinogenicity:	Category 1B IARC:2B EPA:D ACGIH:A4 Japan Society for Occupational Health:2B
Reproductive toxicity:	Not classified
Specific target organ toxicity (Single exposure):	Category 1(respiratory apparatus)
Specific target organ toxicity (Repeated exposure):	Category 1(respiratory apparatus)
Aspiration hazard:	Classification not possible

12. Ecological information

[Chemical product]

Ecotoxicity

Acute hazard:	Category 2
Fish:	LC50 (96hr): 1.1 mg/L[Cyprinodon variegatus] LC50 (96h): 2.11 mg/L[Pimephales promelas] LC50 (96h): 1.81 mg/L[Oncorhynchus mykiss]
Crustacea:	EC50 (48h): 2.2 mg/L[Daphnia magna]
Algae or other aquatic plants:	EC50 (72h): 15 mg/L[Desmodesmus] EC50 (96h): <= 46.78 mg/L[Pseudokirchneriella subcapitata]
Chronic hazard:	Category 2
Fish:	No data
Crustacea:	NOEC (21day): 0.36 mg/L[Daphnia pulex]
Algae or other aquatic plants:	No data

Persistence and degradability: Not rapidly biodegradable

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Bioaccumulative potential:	Low bioaccumulation BFC 3.162
Mobility in soil:	No data
Hazardous to the ozone layer:	Classification not possible

13. Disposal considerations

Disposal methods: When waste material and waste water are to be treated, incinerate them or collect them into specified containers and entrust the disposal to a licensed industrial waste disposal company.

Do not use the used containers for other purposes like filling other substances. After treating the content according to the above description, the empty containers should be reused, recycled, or disposed through suitably qualified or licensed contractor, in accordance with national/local regulations.

The waste water can be treated also by an activated sludge method.

14. Transport information

ICAO/IATA	UN classification :	Class3
	UN number :	1919
	Proper shipping name :	METHYL ACRYLATE, STABILIZED
IMDG	Packing group :	II
	UN classification :	Class3
	UN number :	1919
	Proper shipping name :	METHYL ACRYLATE, STABILIZED
	Packing group :	II
	Marine pollutant:	Yes
Transport regulations:	Transport this product in accordance with local/regional/national/international regulations.	
Special precautions:	At the time of discharge from a tank truck, be sure to ground to prevent static electricity charge and to completely empty the remaining liquid in the pipes. Load the containers in such a way as not to fall down, tumble, or being damaged. Cover the loaded cargo to prevent direct sunlight. (Reference information) Heat accumulation storage test (UN recommendations) SAPT > 50°C (SAPT: Self-accelerating polymerization temperature)	

15. Regulatory information

Inventories

Japan (ENCS) :	Listed
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Korea (KECL) :	Listed
Australia (AIIC) :	Listed
Canada (DSL) :	Listed
China (IECSC) :	Listed
EU (EINECS) :	Listed
New Zealand (NZIoC) :	Listed
USA (TSCA) :	Listed
Philippines (PICCS) :	Listed
Taiwan (TCSI) :	Listed

Restriction

at export (Japan) : Not applicable

16. Other information

Disclaimer:

•The described information is based on the latest material, information, and other data available at the time of preparation; however, it does not intend to guarantee anything like contents, physical/chemical properties, and danger/hazards. Also, the precautions mentioned above are for ordinary handling; therefore, for special handling, it is the obligation of each user of the product to take proper measures suitable for each handling condition. Take enough care for such matters.

•This SDS is based on JIS Z7252: 2019 and JIS Z7253: 2019. If "Not classified" is provided in the GHS classification in Section 2, it means that sufficient information is available to classify the product and sufficient evidence is found to determine that it is not applicable to any hazard categories defined by JIS Z7252. And another case is that it is based on comprehensive judgment by experts.

•Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.